

CHAPTER 8

Aggregate Planning in a Supply Chain

 Instructor

Prof. Ho Young Jeong

 Session

Week 7

Chapter 07

이전 강의 recap

Demand Forecasting in a Supply Chain

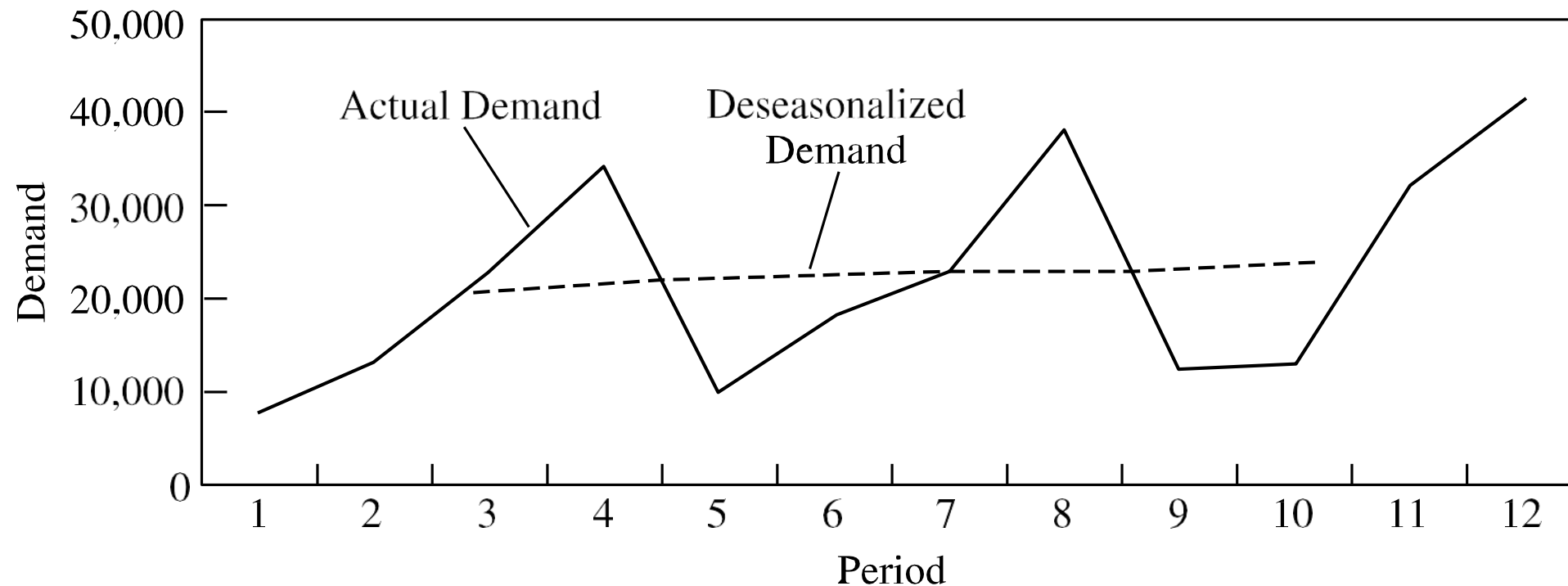


8.3. Aggregate Planning Using Linear Programming

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Forecasting methods for demand prediction and trend analysis

This section covers various time-series forecasting methods including moving average, exponential smoothing, and adaptive forecasting techniques for demand prediction.



Note: Time-series forecasting uses historical data to predict future demand patterns and trends.

8.3. Aggregate Planning Using Linear Programming

Tahoe Salt Co.: Deseasonalizing demand

Example calculation for $t=3$ with $p=4$ (even case) and comparison with $p=3$ (odd case)

	A	B	C	D	E	F	G
1	Tahoe Salt Co. Quarterly Demand						
2							
3	Year	Quarter	Period t	Demand Dt	Season ID	Season mapping note	Decentralized demand
4	1	Q2	1	8,000	1	Q2 -> Season 1	
5	1	Q3	2	13,000	2	Q3 -> Season 2	
6	1	Q4	3	23,000	3	Q4 -> Season 3	19750
7	2	Q1	4	34,000	4	Q1 -> Season 4	20625
8	2	Q2	5	10,000	1		21250
9	2	Q3	6	18,000	2		21750
10	2	Q4	7	23,000	3		22500
11	3	Q1	8	38,000	4		22125
12	3	Q2	9	12,000	1		22625
13	3	Q3	10	13,000	2		24125
14	3	Q4	11	32,000	3		
15	4	Q1	12	41,000	4		

=(D4+D8+2*SUM(D5:D7))/8			
D	E	F	G
Tahoe Salt Co. Quarterly Demand			
t	Demand Dt	Season ID	Season mapping note
1	8,000	1	Q2 -> Season 1
2	13,000	2	Q3 -> Season 2
3	23,000	3	Q4 -> Season 3
4	34,000	4	Q1 -> Season 4
5	10,000	1	
6	18,000	2	

÷ For $t=3, p=4$ (even) Quarterly data, even periodicity

- 1 $D_3 = [D_{3-2} + D_{3+2} + \sum D_i] / 2p$
- 2 $D_3 = [D_1 + D_5 + 2D_2 + 2D_3 + 2D_4] / 8$
- 3 $D_3 = [8,000 + 10,000 + 2*13,000 + 2*23,000 + 2*34,000] / 8$
- 4 $D_3 = 158,000 / 8 = 19,750$

% For $t=3, p=3$ (odd) Quarterly data, odd periodicity

- 1 $D_3 = \sum D_i / p$
- 2 $D_3 = [D_2 + D_3 + D_4] / 3$
- 3 $D_3 = [13,000 + 23,000 + 34,000] / 3$
- 4 $D_3 = 70,000 / 3 = 23,333$

8.3. Aggregate Planning Using Linear Programming

Forecasting methods for demand prediction and trend analysis

Tahoe Salt Co.: Deseasonalizing demand (estimating L and T)

	A	B	C	D	E	F	G	H	I	J
1	요약 출력									
2										
3	회귀분석 통계량									
4	다중 상관	0.958065								
5	결정계수	0.917889								
6	조정된 결	0.904204								
7	표준 오차	414.5033								
8	관측수	8								
9										
10	분산 분석									
11		자유도	제곱합	제곱 평균	F 비	유의한 F				
12	회귀	1	11523810	11523810	67.07182	0.000179				
13	잔차	6	1030878	171813						
14	계	7	12554688							
15										
16		계수	표준 오차	t 통계량	p-값	하위 95%	상위 95%	하위 95.0%	상위 95.0%	
17	Y 절편	18438.99	440.8087	41.82991	1.25E-08	17360.37	19517.61	17360.37	19517.61	
18	X 1	523.8095	63.95925	8.189738	0.000179	367.3069	680.3122	367.3069	680.3122	
19										

$$T = \frac{\sum t \bar{D}_t - n \bar{\bar{D}}}{\sum t^2 - n \bar{t}^2}$$

$$L = \bar{\bar{D}} - T \bar{t}$$

In the example, L = 18,439 and T = 524, so $\bar{D}_t = 18,439 + 524t$

$$\bar{S}_t = D_t / \bar{D}'_t$$

8.3. Aggregate Planning Using Linear Programming

Tahoe Salt Co.: Estimating seasonal factors

Calculating seasonal factors by averaging deseasonalized demand across cycles



Seasonal Factor Calculation

Averaging method for seasonal indices

$$S_i = \sum_{j=0}^{r-1} \bar{S}_{j \cdot p + i} / r$$

Where r = number of seasonal cycles

S_i = Average of all seasonal factors for season i



The overall seasonal factor for a "season" is obtained by averaging all factors for that season across all cycles.



Calculated Seasonal Factors For Tahoe Salt Co. ($p=4$, $r=3$ cycles)

Quarter	Season	Factor	Value
Q2	S1	$(0.42+0.47+0.52)/3$	0.47
Q3	S2	$(0.67+0.83+0.55)/3$	0.68
Q4	S3	$(1.15+1.04+1.32)/3$	1.17
Q1	S4	$(1.66+1.68+1.66)/3$	1.67

	Period t	Demand D_t	Deseasonalized demand $D't$	S_t	Quarter	Season ID	F_t				
3											
4	1	8,000	18963	0.4219	Q2	1			s_1	$=(D4+D8+D12)/3$	
5	2	13,000	19487	0.6671	Q3	2			s_2	0.68	
6	3	23,000	20011	1.1494	Q4	3			s_3	1.17	
7	4	34,000	20535	1.6557	Q1	4			s_4	1.66	
8	5	10,000	21059	0.4749	Q2	1			sum	3.99	
9	6	18,000	21583	0.8340	Q3	2					
10	7	23,000	22107	1.0404	Q4	3					
11	8	38,000	22631	1.6791	Q1	4					
12	9	12,000	23155	0.5182	Q2	1					
13	10	13,000	23679	0.5490	Q3	2					
14	11	32,000	24203	1.3222	Q4	3					
15	12	41,000	24727	1.6581	Q1	4					



Verification

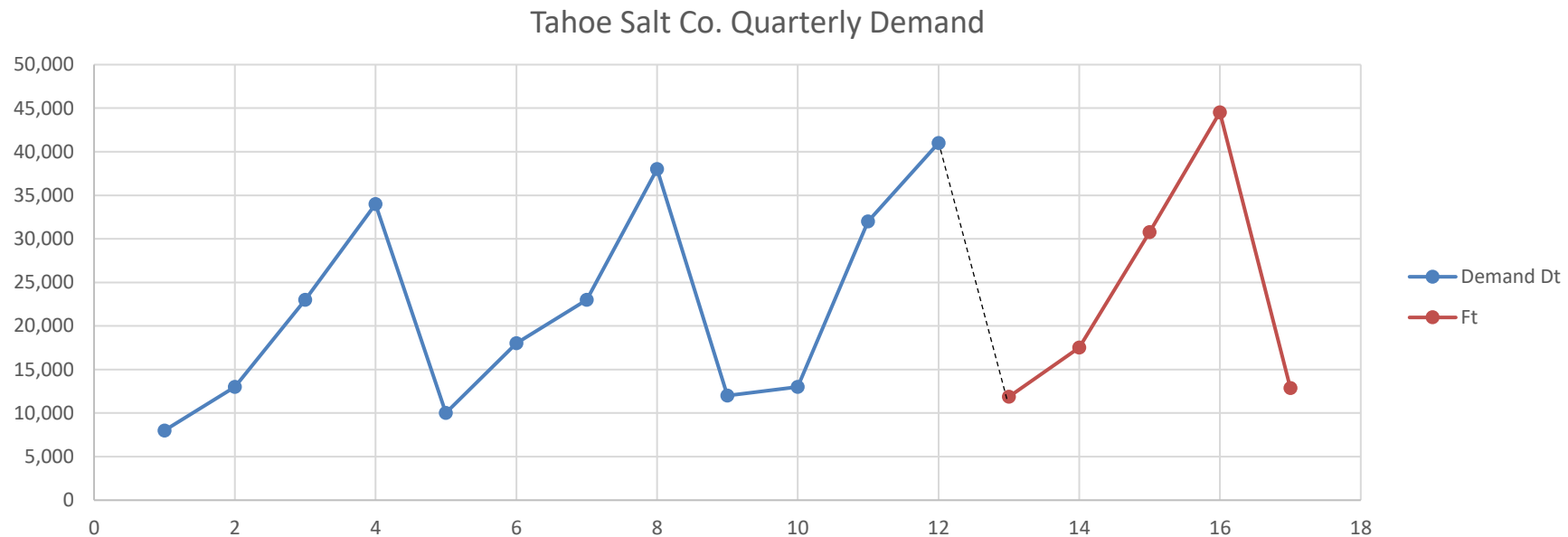
Sum of seasonal factors

Sum of factors: $0.47 + 0.68 + 1.17 + 1.67 = 3.99 \approx 4.0$
(Should equal number of periods in cycle)

8.3. Aggregate Planning Using Linear Programming

Forecasting methods for demand prediction and trend analysis

This section covers various time-series forecasting methods including moving average, exponential smoothing, and adaptive forecasting techniques for demand prediction.



Note: Time-series forecasting uses historical data to predict future demand patterns and trends.

8.3. Aggregate Planning Using Linear Programming

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Simple exponential smoothing method - Exponentially increased weights to recent observations

Exponential Smoothing Formula

The forecast for period $t+1$ is calculated as:

$$F_{t+1} = \alpha D_t + (1-\alpha) F_t$$

Where:

- F_{t+1} = Forecast for next period
- D_t = Actual demand in period t
- α = Smoothing constant ($0 < \alpha < 1$)

Mathematical Proof

Why weights sum to 1:

The sum of all weights is a geometric series: $\alpha + \alpha(1-\alpha) + \alpha(1-\alpha)^2 + \dots = \alpha/(1-(1-\alpha)) = 1$

Exponential Smoothing Concept

$$F_{t+1} = \alpha D_t + (1-\alpha) F_t$$

$$F_t = \alpha D_{t-1} + (1-\alpha) F_{t-1}$$

$$F_{t-1} = \alpha D_{t-2} + (1-\alpha) F_{t-2}$$

$$F_{t+1} = \alpha D_t + \alpha(1-\alpha)D_{t-1} + \alpha(1-\alpha)^2D_{t-2} + \dots + \alpha(1-\alpha)^{t-1}D_1 + (1-\alpha)^tF_1$$

8.3. Aggregate Planning Using Linear Programming

Example 7.3: Holt's model

Japan National Tourist Organization: Visitors forecast to Japan using linear regression for 2002-2007



Problem Statement

Using linear regression for the number of visitors during the year of 2002-2007, with $L_0 = 2,604,842$ and $T_0 = 548,247$. Forecast for year 2008?

Year	Visitors
2002	2,604,842
2003	3,153,089
2004	3,701,336
2005	4,233,360
2006	4,780,000
2007	5,328,247

Level (L_0)

2,604,842

Trend (T_0)

548,247

Smoothing α

0.2 (for level)

Smoothing β

0.1 (for trend)



Forecast Calculation



Using Holt's model with trend correction:

$$F_{t+1} = L_t + T_t$$

$$L_t = \alpha D_t + (1 - \alpha)(L_{t-1} + T_{t-1})$$

$$T_t = \beta (L_t - L_{t-1}) + (1 - \beta) T_{t-1}$$

Forecast for Period 1 (or year 2002):

$$F_1 = L_0 + T_0 = 3,153,809$$

Observed number for Period 1 ($D_1 = 3,417,774$), so

$$E_1 = F_1 - D_1 = -264,685$$

With $\alpha = 0.1$, $\beta = 0.2$,

$$L_1 = \alpha D_1 + (1 - \alpha)(L_0 + T_0) = 3,179,558$$

$$T_1 = \beta (L_1 - L_0) + (1 - \beta) T_0 = 553,541$$

So, $F_2 = L_1 + T_1 = 3,179,558 + 553,541 = 3,733,099$ visitors

Continuing in this manner, F_7 (year 2008) = $L_6 + T_6 = 6,439,353$

For 2008 (t=7):

Chapter Overview & Learning Objectives

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Statistical measures to evaluate forecast accuracy and model performance



Forecast Error

The difference between forecast and actual demand:

$$E_t = F_t - D_t$$

Where E_t = forecast error, F_t = forecast, D_t = actual demand



Mean Squared Error (MSE)

Average of squared forecast errors:

$$MSE = (1/n) \times \sum (E_t)^2$$

Penalizes large errors more heavily



Mean Absolute Deviation (MAD)

Average of absolute forecast errors:

$$MAD = (1/n) \times \sum |E_t|$$

More robust to outliers than MSE



Mean Absolute % Error (MAPE)

Percentage error relative to actual:

$$MAPE = (100/n) \times \sum |E_t/D_t|$$

Useful for comparing across different scales



Bias

Systematic forecast error (should be ~0):

$$Bias = \sum E_t$$

Note: Should fluctuate around 0



Tracking Signal (TS)

Ratio of cumulative error to MAD:

$$TS = Bias / MAD$$

Monitors forecast model adequacy

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의사결정의 빈도(Frequency)와 계획 기간(Planning Horizon)에 따라 3단계로 분류됩니다.



1. 전략 또는 설계

Strategy or Design

장기 (1 ~ 5년)

공급망의 구조 설정 및 자원 배분
(시설 입지, 아웃소싱 여부 등)



2. 계획

Planning

중기 (분기 ~ 1년)

고정된 전략 하에서의 운영 방침 설정
(재고 정책, 수요 예측, 프로모션)



3. 운영

Operation

단기 (일/주 단위)

개별 고객 주문에 대한 실시간 결정
(주문 할당, 피킹, 배송 일정)

💡 이 세 단계의 의사결정은 기업의 성공과 실패를 좌우하는 핵심적인 역할을 수행합니다.

Chapter Overview

- 01** Aggregate Planning and Its Role in a Supply chain
- 02** The Basic Trade-Offs in Aggregate Planning
- 03** Aggregate Planning Using Linear Programming
- 04** Aggregate Planning in Excel

 Supply Chain Management: Strategy, Planning, and Operation (Sunil Chopra, Pearson, 7th edition)

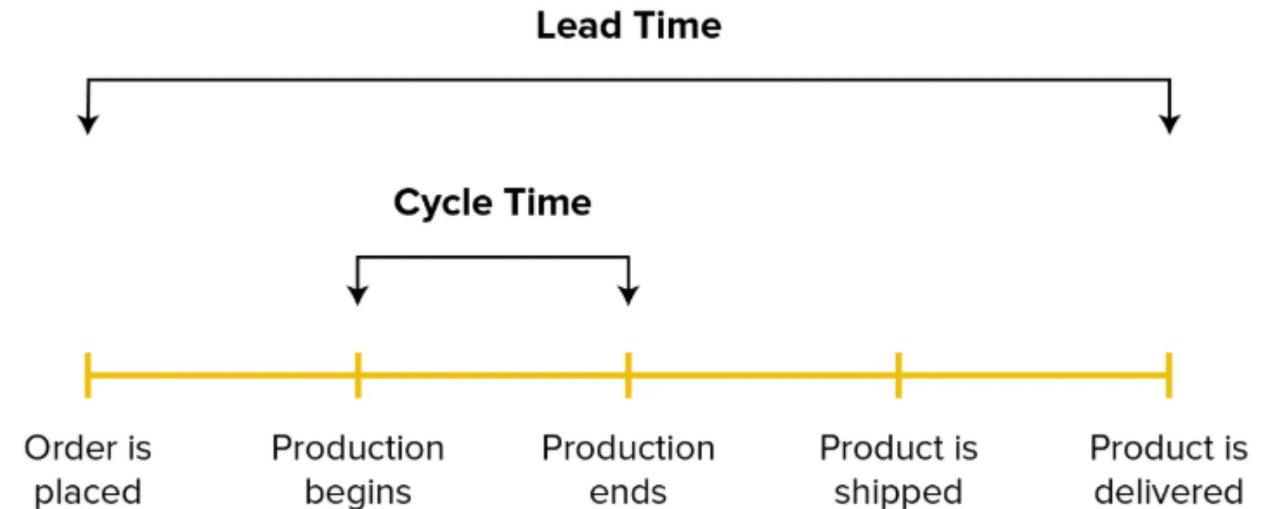
 Lecture Material for Supply Chain Management, by H. Y. Jeong

8.1. Aggregate Planning and Its Role in a Supply chain

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Capacity has a cost, lead times are greater than zero

- Capacity has a cost, lead times are greater than zero
- Given the demand forecast for each period in the planning horizon, aggregate planning determines **the production level, subcontracting level, inventory level, capacity level, and any backlogs for each period** that maximize the firm's profit over the planning horizon
 - Decisions made at a **product family** (not stock-keeping unit: SKU) level
(c.f.) master production schedule (MPS)
 - Time frame of between **3 and 18 months**



8.1. Aggregate Planning and Its Role in a Supply chain

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Aggregate Planning and Its Role in a Supply chain



Aggregate planning requires inputs from all stages of the supply chain, and its output is also of value to both upstream and downstream partners, thus **all supply chain stages should work together** on an aggregate plan that will optimize supply chain performance.



Surprisingly, many companies do not create aggregate plans and instead rely **solely on orders from their customers** to determine their production schedules.



8.1. Aggregate Planning and Its Role in a Supply chain

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Information requirements



Information requirements

- Demand forecast in each period
- \$ Production costs
- ⌚ Labor costs: regular time (\$/hr), overtime (\$/hr)
- 🚚 Subcontracting costs (\$/unit)
- ⚙️ Labor/machine hours required per unit
- 📦 Inventory holding cost (\$/unit/period)
- ⚠️ Lost sales cost(\$/unit) or backlog cost(\$/unit/period)
- 👥 Cost of changing capacity:
hiring or layoff (\$/worker), adding or reducing machine capacity (\$/machine)
- 🔒 **Constraints:**
Limits on overtime and subcontract, layoffs, capital available, stockouts (lost sales or backlogs)

$$\text{Minimize } \sum_{t=1}^6 640W_t + \sum_{t=1}^6 6O_t + \sum_{t=1}^6 300H_t + \sum_{t=1}^6 500L_t \\ + \sum_{t=1}^6 2I_t + \sum_{t=1}^6 5S_t + \sum_{t=1}^6 10P_t + \sum_{t=1}^6 30C_t$$

$$W_t = W_{t-1} + H_t - L_t \quad (W_0 = 80)$$

$$P_t \leq 40W_t + \frac{O_t}{4}$$

$$I_{t-1} + P_t + C_t = D_t + S_{t-1} + I_t - S_t \quad (S_0 = 0, I_0 = 1,000, I_6 \geq 500)$$

$$O_t \leq 10W_t$$

$$S_6 = 0$$

8.1. Aggregate Planning and Its Role in a Supply chain

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Outputs of aggregate planning

Determinations through aggregate planning (per each period)

- **Production quantity from regular time, overtime, and subcontract:** used to determine outsourcing contract
- **Machine capacity (machine capacity increase/decrease):** used to determine whether new production equipment should be purchased or available equipment idled
- **Stockout quantity:** used to estimate customer service levels
- **Workforce size (workforce hired/laid off):** used to determine the labor issues likely to be encountered
- **Inventory held:** used to determine the warehouse space and working capital required



Quality impact:

The quality of aggregate plan has a significant impact on the firm's profitability because the aggregate plan affects the optimal match of supply and demand

8.2. The Basic Trade-Offs in Aggregate Planning

IE31500-SCM

Trade-offs among capacity, inventory, and stockout costs in aggregate planning

Trade-offs among capacity, inventory, and stockout costs in aggregate planning

Three pure strategies



Chase strategy

Using capacity as the lever



Flexibility strategy

Using workforce or machine utilization as the lever



Level strategy

Using inventory and backlogs as the lever



Mixed strategy

A combination of two or more pure strategies

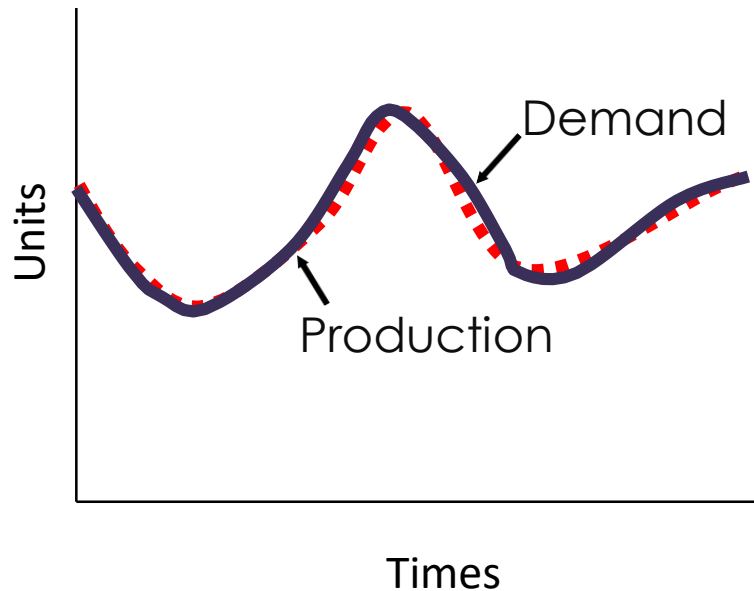


8.2. The Basic Trade-Offs in Aggregate Planning

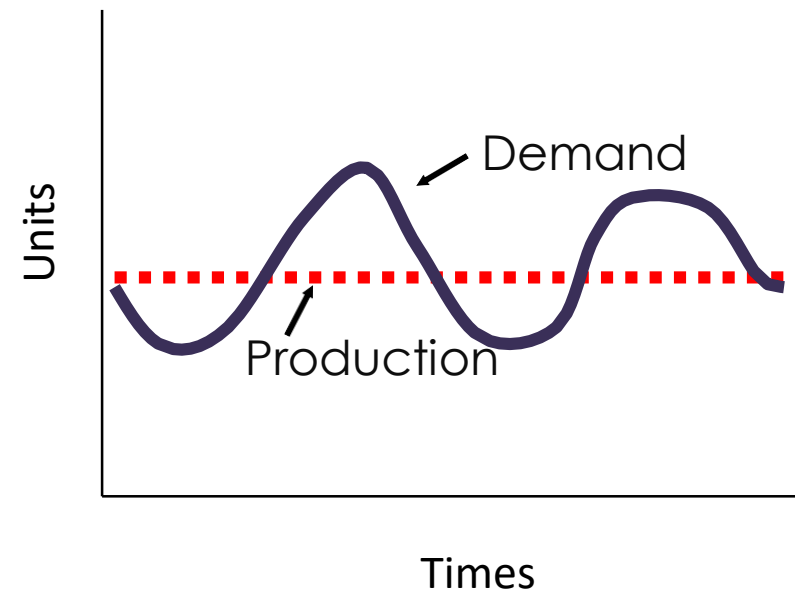
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Chase of Flexibility vs. Level Strategy

Chase or Flexibility strategy



Level strategy



8.2. The Basic Trade-Offs in Aggregate Planning

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Chase strategy

Chase strategy

- Production rate is synchronized with **demand by varying capacity (number of machines or workers)** as the demand rate varies
- However, in practice, it is often difficult to vary capacity on short notice
- Expensive if cost of varying capacity is high
- Negative effect on workforce morale
- Results in very low levels of inventory
- Should be used when inventory holding costs are high and costs of changing capacity are low



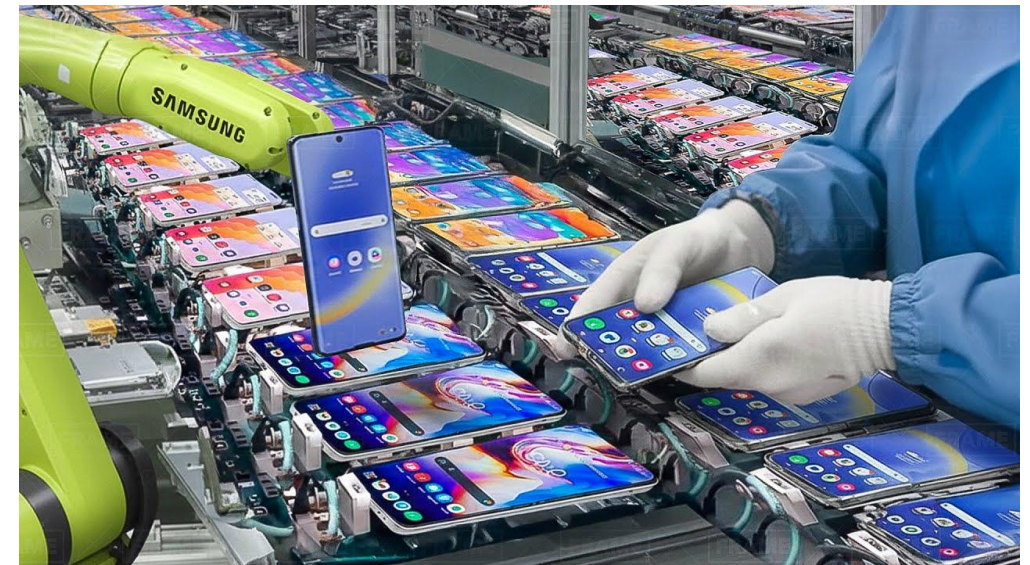
8.2. The Basic Trade-Offs in Aggregate Planning

IE31500-SCM

Flexibility strategy

Flexibility strategy

- Workforce is kept stable, but **the number of hours worked is varied** over or below the regular time to synchronize production with demand
- Avoids morale problems of the chase strategy
- Low levels of inventory but with lower average machine utilization
Should be used when inventory holding costs are
- relatively high and machine capacity is relatively inexpensive



8.2. The Basic Trade-Offs in Aggregate Planning

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Level strategy

Level strategy

- **Maintains stable machine capacity and workforce levels** with a constant output rate
- Inventories are built up in anticipation of future demand or backlogs are carried over from high- to low-demand periods
- Better for worker morale - stable workforce and predictable working conditions
- Large inventories and backlogs may accumulate during low-demand periods
- Should be used when inventory holding and backlog costs are relatively low

Key Insight:

Level strategy prioritizes workforce stability over inventory costs, making it suitable when labor is more valuable than holding inventory.



8.3. Aggregate Planning Using Linear Programming

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Red Tomato Tools: Manufacturer of gardening equipment

Material cost/aggregate unit: $15 \times 0.1 + 7 \times 0.25 + \dots + 13 \times 0.15 = \10

Revenue/aggregate unit: $54 \times 0.1 + 30 \times 0.25 + \dots + 48 \times 0.15 = \40

Net production time/aggregate unit = 4 hours $\times 5.76 = 5.6 + (8/50)$

Product SKU table (A-F products with columns for costs, revenue, setup time, batch size, production time, percentage share)

Product (SKU)	Material Cost / Unit (\$)	Revenue / Unit (\$)	Setup Time / Batch (hour)	Average Batch Size	Production Time / Unit (hour)	Net Production Time / Unit (hour)	Percentage Share of Units Sold
A	15	54	8	50	5.60	5.76	10%
B	7	30	6	150	3.00	3.04	25%
C	9	39	8	100	3.80	3.88	20%
D	12	49	10	50	4.80	5.00	10%
E	9	36	6	100	3.60	3.66	20%
F	13	48	5	75	4.30	4.37	15%

Refer to provided xlsx file

8.3. Aggregate Planning Using Linear Programming

Demand Forecast at Red Tomato Tools

Sets

$t \in \{1, 2, 3, 4, 5, 6\}$

Planning horizon (1: Jan, 2: Feb, 3: Mar, 4: Apr, 5: May, 6: Jun)

Demand Parameters

$D_1 = 1600, D_2 = 3000, D_3 = 3200, D_4 = 3800, D_5 = 2200, D_6 = 2200$

- Highly seasonal demand
- Capacity determined mainly by workforce size (i.e., flexible workforce)
- Subcontracting, inventories, and backlogs are allowed
- Minimizing cost is the same as maximizing profit (Why?)

Demand forecast	
Month	Demand
Jan	1600
Feb	3000
Mar	3200
Apr	3800
May	2200
Jun	2200
Total	16000

Refer to provided xlsx file

8.3. Aggregate Planning Using Linear Programming

IE31500-SCM

Input data for Red Tomato Tools

Inputs	
Materials cost (\$/unit)	10
Inventory holding cost (\$/unit/month)	2
Stockout cost (\$/unit/month)	5
Hiring cost (\$/worker)	300
Layoff cost (\$/worker)	500
Labor hours per unit	4
Regular time cost (\$/hour)	4
Overtime cost (\$/hour)	6
Subcontract cost (\$/unit)	30
Capacity per worker (units/month)	40
Initial workforce W_0	80
Initial inventory I_0	1000
Minimum ending inventory (end of Jun)	500
Max overtime hours per worker per month	10

Cost Parameters

$c_m = 10$: Material cost per unit (\$/unit)

$h = 2$: Inventory holding cost per unit per month (\$/unit/month)

$b = 5$: Stockout or backlog cost per unit per month (\$/unit/month)

$c_H = 300$: Hiring cost per worker (\$/worker)

$c_L = 500$: Layoff cost per worker (\$/worker)

$c_r = 640$: Regular time labor cost per worker per month

(40 units of capacity per month x 4 hours per unit x 4 per hour = \$640/worker)

$c_o = 24$: Overtime labor cost per unit (4 hours per unit x \$6 per hour = \$24/unit)

$c_s = 30$: Subcontracting cost per unit (\$/unit)

$m = 40$: Capacity per worker (unit/month)

Capacity and Initial States

$W_0 = 80$: Initial workforce (workers)

$I_0 = 1000$: Initial inventory level (units)

$I_6 \geq 500$: Minimum required ending inventory at the end of month 6 (June)

$O_{\max} = 10$: Maximum overtime hours per worker per month (hours)

Refer to provided xlsx file

8.3. Aggregate Planning Using Linear Programming

Decision variables

Decision variables

Decision variables		
# of Workforce in period t	W_t	$640 \cdot W_t$
Overtime hours in period t	O_t	$6 \cdot O_t$
# of newly hired in period t	H_t	$300 \cdot H_t$
# of layoff in period t	L_t	$500 \cdot L_t$
# of Inventories in period t	I_t	$2 \cdot I_t$
# of stocking out in period t	S_t	$5 \cdot S_t$
# of production units in period t	P_t	$10 \cdot P_t$
# of subcontract units in period t	C_t	$30 \cdot C_t$

Refer to provided excel file

8.3. Aggregate Planning Using Linear Programming

Linear programming model

$$\text{Minimize } \sum_{t=1}^T (c_m P_t + c_r W_t + c_o O_t + c_s C_t + h I_t + b S_t + c_H H_t + c_L L_t) \quad \text{Minimization of the total cost}$$

$$I_{t-1} - S_{t-1} + P_t + \frac{O_t}{k} + C_t = D_t + I_t - S_t \quad \forall t \quad \text{Inventory Balance Constraint}$$

$$W_t = W_{t-1} + H_t - L_t \quad \forall t \quad \text{Workforce Balance Constraint}$$

$$P_t \leq m \cdot W_t + \frac{O_t}{k} \quad \forall t \quad \text{Production Capacity Constraint}$$

$$O_t \leq O_{max} \cdot W_t \quad \forall t \quad \text{Overtime Hours Limit Constraint}$$

$$I_T \geq I_{min} \quad \text{Ending Inventory Requirement}$$

8.3. Aggregate Planning Using Linear Programming

Linear programming model

- **Objective function:** Minimization of the total cost (regular- time labor cost + overtime labor cost + cost of hiring and layoffs + cost of holding inventory + cost of stocking out + Material cost + cost of subcontracting) incurred during the planning horizon

$$\begin{aligned} \text{Minimize } & \sum_{t=1}^6 640W_t + \sum_{t=1}^6 6O_t + \sum_{t=1}^6 300H_t + \sum_{t=1}^6 500L_t \\ & + \sum_{t=1}^6 2I_t + \sum_{t=1}^6 5S_t + \sum_{t=1}^6 10P_t + \sum_{t=1}^6 30C_t \end{aligned}$$

Refer to provided xlsx file

8.3. Aggregate Planning Using Linear Programming

IE31500-SCM

Linear programming model

- **Constraints**

$$W_t = W_{t-1} + H_t - L_t \quad (W_0 = 80)$$

- Workforce size for each month is based on hiring and layoffs

$$P_t \leq 40W_t + \frac{O_t}{4}$$

- Production for each month cannot exceed capacity

$$I_{t-1} + P_t + C_t = D_t + S_{t-1} + I_t - S_t \quad (S_0 = 0, I_0 = 1,000, I_6 \geq 500)$$

- Inventory balance for each month

$$O_t \leq 10W_t$$

- Overtime for each month

$$S_6 = 0$$

- No more stockout after June.

Refer to provided xlsx file

8.3. Aggregate Planning Using Linear Programming

Solved using “Solver” in Excel

Decision variables (blue) and constraint checks															
t	Month	Hired Ht	Laid off Lt	Workfor ce Wt	Overtim e hours Ot	Inventory It	Stockout St	Subcontr act Ct	Productio n Pt	Demand Dt	Capacity (units)	Cap slack (>=0)	Workforce balance (=0)	Inventory balance (=0)	OT limit (<=0)
1	Jan	0	16	64	0	1960	0	0	2560	1600	2560	0	0	0	-640
2	Feb	0	0	64	0	1520	0	0	2560	3000	2560	0	0	0	-640
3	Mar	0	0	64	0	880	0	0	2560	3200	2560	0	0	0	-640
4	Apr	0	0	64	0	0	220	140	2560	3800	2560	0	0	0	-640
5	May	0	0	64	0	140	0	0	2560	2200	2560	0	0	0	-640
6	Jun	0	0	64	0	500	0	0	2560	2200	2560	0	0	0	-640
		0	16	384	0	5000	220	140	15360	16000	15360	0	0	0	-3840

8.3. Aggregate Planning Using Linear Programming

IE31500-SCM

Solved using “Solver” in Excel

Objective (Minimize total cost)

Regular-time labor	\$245,760
Overtime labor	\$0
Hiring	\$0
Layoff	\$8,000
Inventory holding	\$10,000
Stockout	\$1,100
Materials (production)	\$153,600
Subcontract	\$4,200
TOTAL COST	\$422,660

8.3. Aggregate Planning Using Linear Programming

Example 8.1: Impact of higher demand variability

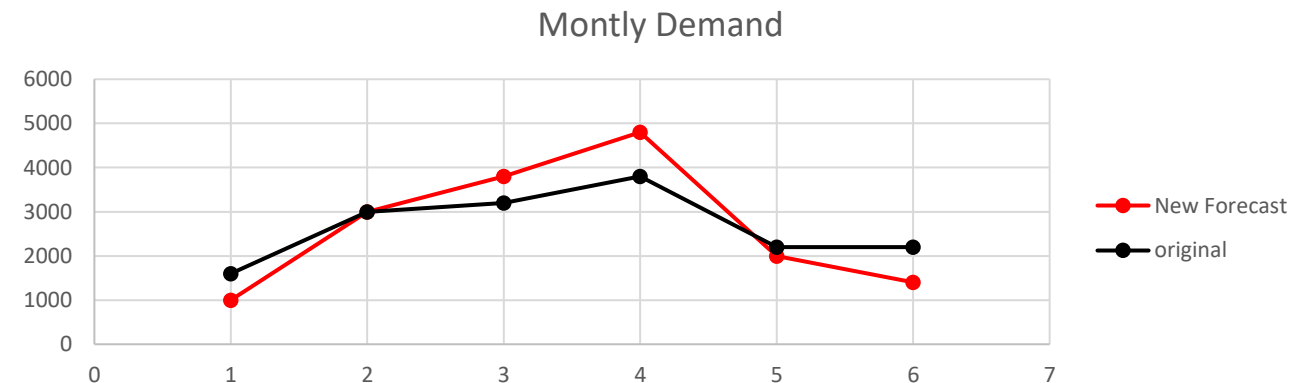
Original

Demand forecast	
Month	Demand
Jan	1600
Feb	3000
Mar	3200
Apr	3800
May	2200
Jun	2200
Total	16000

New

Demand forecast	
Month	Demand
Jan	1000
Feb	3000
Mar	3800
Apr	4800
May	2000
Jun	1400
Total	16000

- Increased demand fluctuation
- Overall demand is same (16,000 units)



8.3. Aggregate Planning Using Linear Programming

Example 8.1: Impact of higher demand variability

Decision variables (blue) and constraint checks															
t	Month	Hired H _t	Laid off L _t	Workforce W _t	Overtime hours O _t	Production P _t	Subcontract C _t	Inventory I _t	Stockout S _t	Demand D _t	Capacity (units)	Cap slack (≥0)	Workforce balance (=0)	Inventory balance (=0)	OT limit (≤0)
1	Jan	0	16	64	0	2560	0	2560	0	1000	2560	0	0	0	-640
2	Feb	0	0	64	0	2560	0	2120	0	3000	2560	0	0	0	-640
3	Mar	0	0	64	0	2560	0	880	0	3800	2560	0	0	0	-640
4	Apr	0	0	64	0	2560	140	0	1220	4800	2560	0	0	0	-640
5	May	0	0	64	0	2560	0	0	660	2000	2560	0	0	0	-640
6	Jun	0	0	64	0	2560	0	500	0	1400	2560	0	0	0	-640

8.3. Aggregate Planning Using Linear Programming

IE31500-SCM

Example 8.1: Impact of higher demand variability

Original

Objective (Minimize total cost)	
Regular-time labor	\$245,760
Overtime labor	\$0
Hiring	\$0
Layoff	\$8,000
Inventory holding	\$10,000
Stockout	\$1,100
Materials	
(production)	\$153,600
Subcontract	\$4,200
TOTAL COST	\$422,660

New

Objective (Minimize total cost)	
Regular-time labor	\$245,760
Overtime labor	\$0
Hiring	\$0
Layoff	\$8,000
Inventory holding	\$12,120
Stockout	\$9,400
Materials	
(production)	\$153,600
Subcontract	\$4,200
TOTAL COST	\$433,080

- Total cost over planning horizon = \$433,080 > \$422,660
(due to increased demand fluctuation)

8.3. Aggregate Planning Using Linear Programming

Example 8.2: Impact of lower costs of hiring and layoff to \$50 each

Original

Hiring cost (\$/worker) 300
Layoff cost (\$/worker) 500

New

Hiring cost (\$/worker) 50
Layoff cost (\$/worker) 50

Decision variables (blue) and constraint checks															
t	Month	Hired Ht	Laid off Lt	Workf orce Wt	Overti me hours Ot	Produc tion Pt	Subcontr act Ct	Invent ory It	Stock out St	Dema nd Dt	Capacit y (units)	Cap slack (>=0)	Workfor ce balance (=0)	Inventor y balance (=0)	OT limit (<=0)
1	Jan	0	35	45	0	1800	0	1200	0	1600	1800	0	0	0	-450
2	Feb	0	0	45	0	1800	0	0	0	3000	1800	0	0	0	-450
3	Mar	42	0	87	0	3480	0	280	0	3200	3480	0	0	0	-870
4	Apr	1	0	88	0	3520	0	0	0	3800	3520	0	0	0	-880
5	May	0	27	61	0	2440	0	240	0	2200	2440	0	0	0	-610
6	Jun	0	0	61	0	2440	20	500	0	2200	2440	0	0	0	-610

8.3. Aggregate Planning Using Linear Programming

IE31500-SCM

Example 8.2: Impact of lower costs of hiring and layoff to \$50 each

Original

Hiring cost (\$/worker)	300
Layoff cost (\$/worker)	500

Objective (Minimize total cost)

Regular-time labor	\$245,760
Overtime labor	\$0
Hiring	\$0
Layoff	\$8,000
Inventory holding	\$12,120
Stockout	\$9,400
Materials (production)	\$153,600
Subcontract	\$4,200
TOTAL COST	\$433,080

New

Hiring cost (\$/worker)	50
Layoff cost (\$/worker)	50

Objective (Minimize total cost)

Regular-time labor	\$247,680
Overtime labor	\$0
Hiring	\$2,150
Layoff	\$3,100
Inventory holding	\$4,440
Stockout	\$0
Materials (production)	\$154,800
Subcontract	\$600
TOTAL COST	\$412,770

8.3. Aggregate Planning Using Linear Programming

Rough master production schedule at Red Tomato for Period 1

- Aggregate production quantity for Period 1= 2,560 units
- Percentage share units sold of six products:
 - A = 10%, B = 25%, C = 20%, D = 10%, E = 20%, F = 15%

Product (SKU)	Material Cost / Unit (\$)	Revenue / Unit (\$)	Setup Time / Batch (hour)	Average Batch Size	Production Time / Unit (hour)	Net Production Time / Unit (hour)	Percentage Share of Units Sold	Production quantity	Number of Setups	Setup Time	Production Time (hours)
A	15	54	8	50	5.60	5.76	0.10	256.00	5.00	40.00	1433.60
B	7	30	6	150	3.00	3.04	0.25	640.00	4.00	24.00	1920.00
C	9	39	8	100	3.80	3.88	0.20	512.00	5.00	40.00	1945.60
D	12	49	10	50	4.80	5.00	0.10	256.00	5.00	50.00	1228.80
E	9	36	6	100	3.60	3.66	0.20	512.00	5.00	30.00	1843.20
F	13	48	5	75	4.30	4.37	0.15	384.00	5.00	25.00	1651.20

production
quantity for
Period 1
(unit)

2560

8.3. Aggregate Planning Using Linear Programming

Forecast error and aggregate plans

Forecast error and aggregate plans: Flexibility must be built into the aggregate plan to satisfy demand that is higher than forecast. Ways to create a buffer for forecast error using safety inventory or safety capacity.

Ways to create a buffer for forecast error:

- **Build and carry extra inventories**
 - Create safety stock to buffer against demand fluctuations
- **Use overtime**
 - Increase capacity temporarily during peak demand periods
- **Carry extra workforce permanently**
 - Maintain additional staff to handle demand variability
- **Use subcontractors**
 - Outsource production to external partners when needed
- **Purchase capacity or product from a spot (open) market**
 - Buy additional capacity or products from external markets

Thank You!

경청해 주셔서 감사합니다!



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